

Week 9
Discussion lab 8
Machine Learning



Sun Mar 9, 2025



A4

due by 11:59pm



Checkpoint #2: EDA

due by 11:59pm



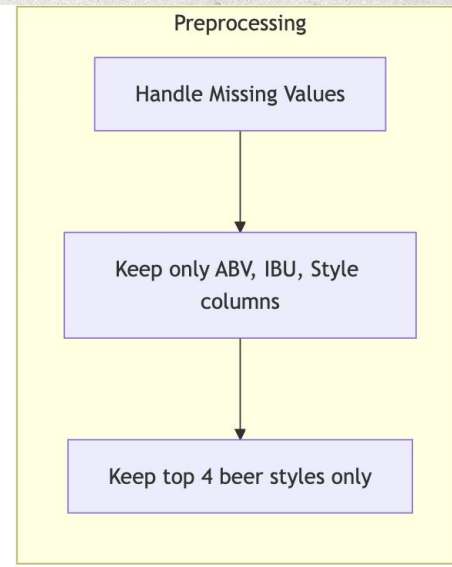
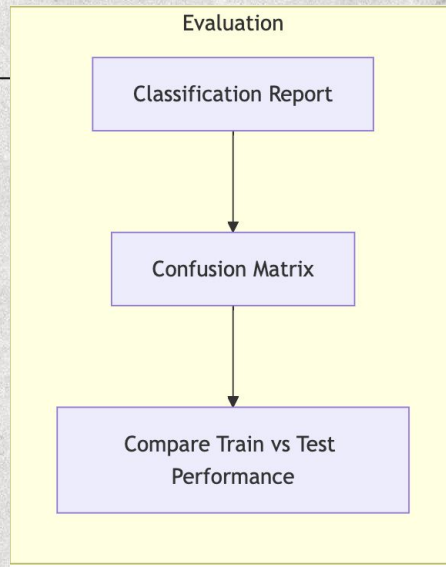
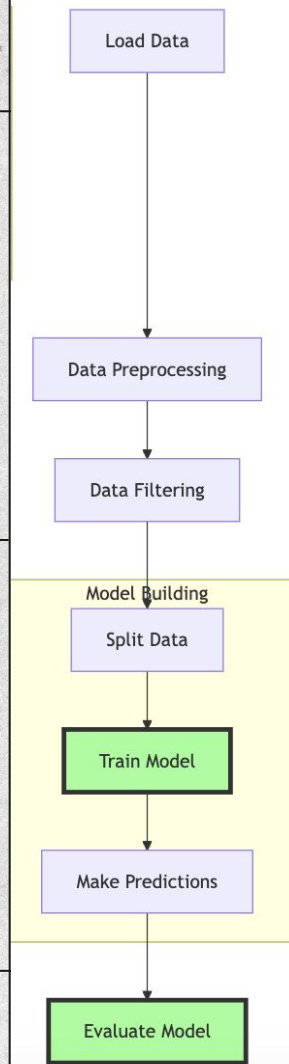
D8

due by 11:59pm



Q9

due by 11:59pm



1. Data preprocessing: Handle missing values and extract ABV, IBU features
2. Filter data to keep only top 4 most common beer styles
3. Split data into training (80%) and test (20%) sets
4. Train SVM model and generate predictions
5. Evaluate model using classification reports and confusion matrices for both training and test sets

Part I: Data, Wrangling, & EDA



1. Analyze missing values(`.isnull().sum(axis=0)`)

	Name	ABV	IBU		Name	ABV	IBU		Name	1
0	Beer1	5.0	45.0	0	False	False	False		ABV	1
1	Beer2	NaN	60.0	1	False	True	False		IBU	2
2	Beer3	7.5	NaN	2	False	False	True			
3	None	4.8	NaN	3	True	False	True			

dtype: int64

2. Remove rows with missing values in style, abv, ibu(`dropna(subset=[])`)
3. Merge beer and brewery datasets(left join)
 - Why left join?
 - How does left join works?
4. Filter dataset to keep only top 4 styles (`.value_counts()[:].index.tolist()`)



Part II : Prediction Model



1. Extract features (X: ABV, IBU) and labels (Y: Style)

```
data_x = beer_df[['abv','ibu']]
```

```
data_y= np.array(beer_df['style'])
```

2. Split data into train/test sets

```
train_X = data_x[:num_training]
```

```
train_Y = data_y[num_training:]
```

```
test_X = data_x[num_training:]
```

```
test_Y = data_y[num_training:]
```

3. Train SVM model and generate predictions

```
beer_clf = train(train_X, train_Y)
```

```
beer_clf.predict(train_X)
```

```
beer_clf.predict(test_X)
```



Part III : Model Assessment



1. Generate classification reports (precision, recall, f1-score)
2. Create confusion matrices
3. Compare training vs testing performance
Train accuracy vs test accuracy
4. Analyze where model performs well/poorly
F1 score
5. Evaluate potential overfitting



Thanks!

Any questions?